



PalmWatch

WHITE PAPER / VERSION 1.0

Evidence-led oil-palm intelligence

A proposed research and operating framework for connecting calibrated aerial evidence, palm-level history, and field confirmation.

Designed for estate teams, agronomists, and research partners

Screening supports expert review; it does not replace field or laboratory confirmation.

September 2026 | [palmwatch site](#)

WHY PALMWATCH

A continuous record for every palm.

PalmWatch is being developed as a field-research platform that turns repeat aerial surveys into a transparent review queue for expert inspection.

01 Identity before inference

Every mapped palm receives a stable identifier, baseline, and time-linked observation history.

02 Signals in agreement

Spectral, structural, thermal, temporal, and contextual evidence are reviewed together.

03 Verification closes the loop

Field findings and, when required, laboratory confirmation remain attached to the record.

The first model target is a three-way screen: healthy, other stress, and suspected Ganoderma. More detailed labels require sufficient confirmed samples.

MINIMUM DEFENSIBLE SCOPE

Scale the evidence before scaling the claim.

These figures are proposed targets for an expanded NBL study. They are not deployment results, accuracy claims, or completed observations.

120

balanced study farms

6,000

mapped palms

3

repeat survey rounds

18,000

tree observations

Design guardrails

- Balance farms across geography, planting age, management, and observable stress contexts.
- Separate training, validation, and locked testing by farm - never randomly by tree.
- Repeat surveys to distinguish persistent decline from seasonal or flight-condition noise.

SIGNAL FAMILIES

Useful screening begins when evidence agrees.

No universal vegetation-index threshold can diagnose a palm. PalmWatch combines calibrated measurements, local comparison, temporal change, and confirmed field labels.

01

Spectral

NDVI, NDRE, GNDVI, red edge, NIR, and calibrated reflectance.

02

Structural

Crown area, density, symmetry, visible gaps, yellowing, and change over time.

03

Thermal

Canopy temperature, transpiration stress, and neighbour-relative anomalies.

04

Contextual

Soil, weather, age, seed, irrigation, fertilizer, infestation, and management history.

A flagged palm is a prioritised observation, not a diagnosis.

FIVE CONNECTED SERVICES

From map to monitored outcome.

Each service produces a traceable hand-off rather than an isolated dashboard event.

01

Map & inventory

Estate boundaries, blocks, and stable palm identities.

02

Calibrated survey

Repeat RGB, multispectral, and thermal capture under field protocols.

03

Risk screening

Palm history and relevant neighbours reveal persistent change.

04

Field verification

Agronomists inspect prioritised palms and attach findings.

05

Monitor & report

Follow-ups connect intervention, outcome, and estate-level reporting.

RESEARCH DISCIPLINE

Build evidence that can survive a new farm.

Validation should test whether the approach generalises across locations, seasons, and management contexts - not whether it remembers a familiar plantation.

Farm-level split

Keep palms from one farm in only one of training, validation, or locked testing.

Repeat capture

Use multiple survey rounds to test persistence and reduce one-flight noise.

Confirmed labels

Retain expert observations and laboratory confirmation where required.

Transparent review

Show the contributing signal families and relevant comparison context.

Initial evaluation questions

Can the screen separate healthy palms, other stress, and suspected Ganoderma? Does performance hold on farms never seen during training? Are prioritised cases explainable and operationally reviewable?

WHAT TEAMS RECEIVE

Decision support with an audit trail.

The platform is designed to keep the estate view connected to the palm, observation, assignment, and field outcome.



- Palm identity and survey history
- Explainable review queue
- Field assignments and notes
- Repeat-monitoring reports

Limitations stated plainly

PalmWatch screens for elevated risk; field or laboratory confirmation remains essential.

Thresholds and model behaviour must be calibrated to the study population and capture protocol.

Detailed disease or severity labels are only defensible when confirmed sample counts support them.

NEXT STEP

Move from a visual demo to a validated field study.

A PalmWatch pilot should begin with protocol agreement, balanced sampling, farm-level data separation, and a field-confirmation workflow.

White paper scope

This document consolidates the current PalmWatch product narrative, proposed study targets, evidence architecture, service workflow, validation rules, outputs, and limitations. It contains no deployment performance or diagnostic-accuracy claim.